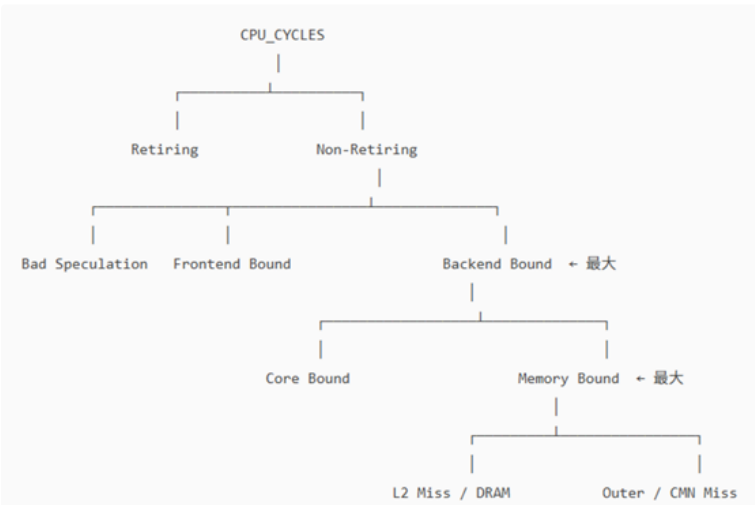
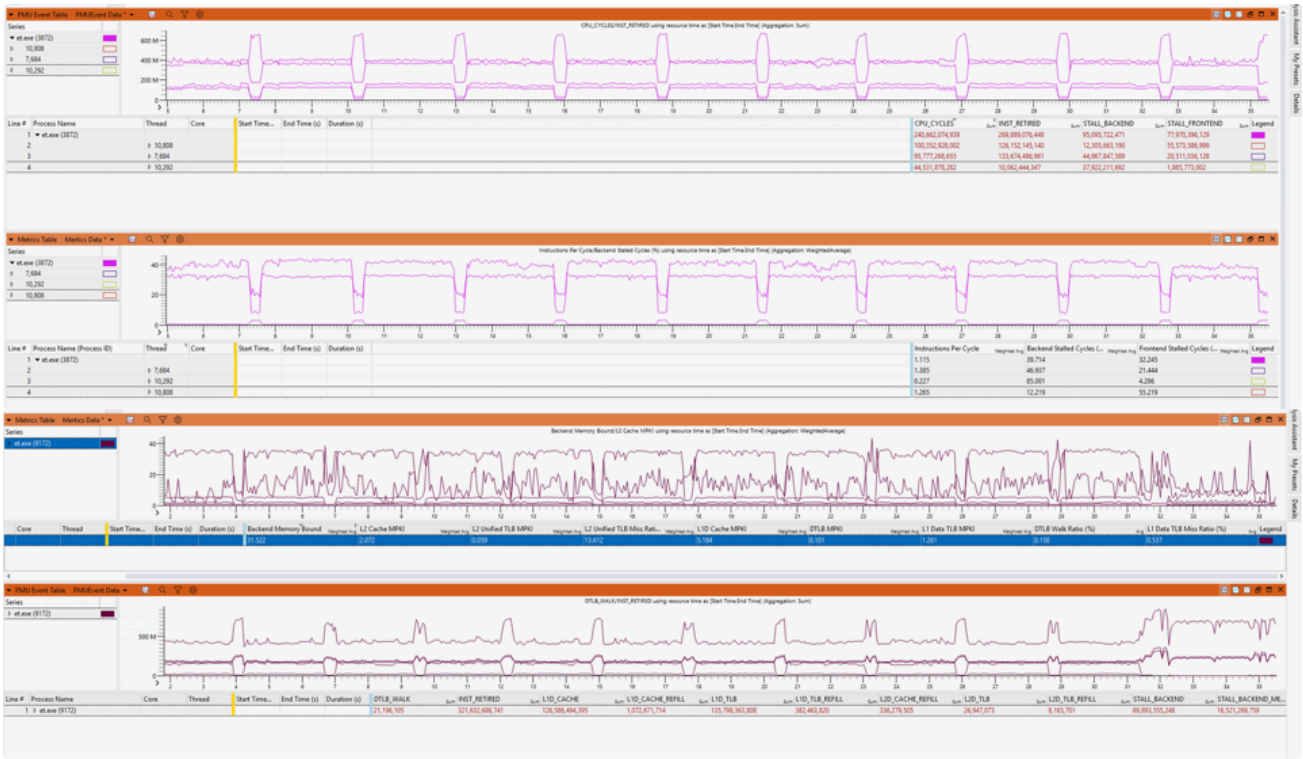
WPS Producer–Consumer Architecture Optimization

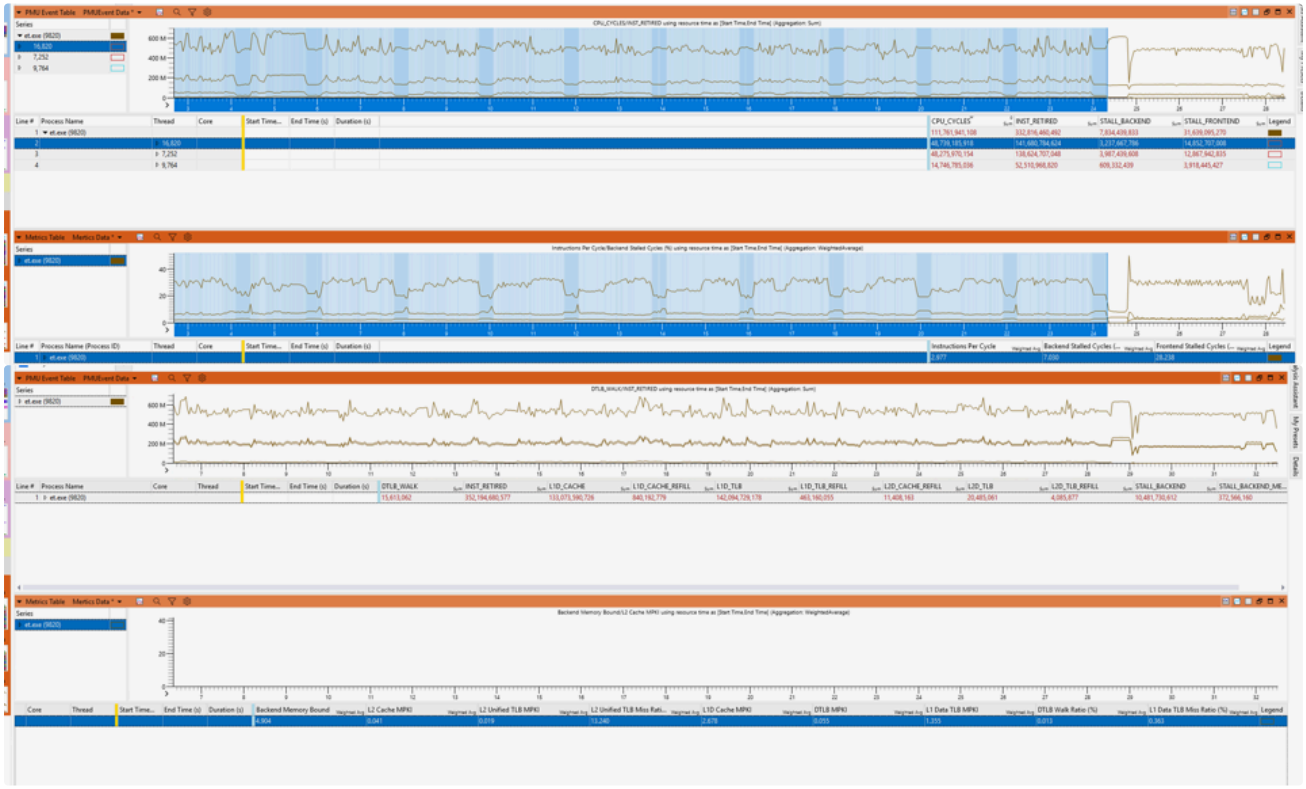
1. WPS uses a producer–consumer architecture when opening large Excel files.
2. Due to the scheduler behavior on WoS, these two threads are sometimes placed on different clusters (cross-L2) to achieve higher boost frequencies.
   core4(cluster1) and core8(cluster2)

3. This scheduling strategy works well for most applications, but WPS relies heavily on a producer–consumer model with intensive access to volatile variables.



4. As a result, cross-cluster execution triggers significant inter-core communication overhead, causing data to move between L2 → L2 or L2 → SLC, which introduces substantial latency.
5. The performance metrics clearly show increased backend memory bound, primarily caused by L2 cache misses.
   Stall_Backend = 39.7%, Backend_Memory_Bound=31.5%, L2_Cache_MPKI =2.07



6. After placing the producer and consumer threads on the same L2 cluster, we observe a notable reduction in both memory bound and L2 cache misses, confirming that co-locating the threads significantly improves performance.

   Stall_Backend = 7.0%, Backend_Memory_Bound=4.9%, L2_Cache_MPKI =0.04